

PRAKTIKUM BASIS DATA

MODUL 9

Scripting Procedure dan Function



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Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

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Code

Query :

```
CREATE TABLE hrememployees (
  employee_id      NUMBER(6)      NOT NULL,
  first_name       VARCHAR2(20),
  last_name        VARCHAR2(25)   NOT NULL,
  email            VARCHAR2(25)   NOT NULL,
  phone_number     VARCHAR2(20),
  hire_date        DATE           NOT NULL,
  job_id           VARCHAR2(10)   NOT NULL,
  salary           NUMBER(8,2),
  commission_pct   NUMBER(2,2),
  manager_id       NUMBER(6),
  department_id    NUMBER(4),
  CONSTRAINT emp_emp_id_pk PRIMARY KEY (employee_id)
);

-- INSERT data untuk department_id = 90 (Executive - dipakai soal no.4)
INSERT INTO hrememployees VALUES (100, 'Steven', 'King', 'SKING', '515.123.4567', TO_DATE('17-JUN-20
'AD_PRES', 24000, NULL, NULL, 90);
INSERT INTO hrememployees VALUES (101, 'Neena', 'Kochhar', 'NKOCHHAR', '515.123.4568', TO_DATE('21-SEP-20
17000, NULL, 100, 90);
INSERT INTO hrememployees VALUES (102, 'Lex', 'De Haan', 'LDEHAAN', '515.123.4569', TO_DATE('13-JAN-20
17000, NULL, 100, 90);

-- INSERT tambahan department lain (agar SELECT * tidak kosong)
INSERT INTO hrememployees VALUES (103, 'Alexander', 'Hunold', 'AHUNOLD', '590.423.4567', TO_DATE('03-JAN-2
9000, NULL, 102, 60);
INSERT INTO hrememployees VALUES (104, 'Bruce', 'Ernst', 'BERNST', '590.423.4568', TO_DATE('21-MAY-2
'IT_PROG', 6000, NULL, 103, 60);
INSERT INTO hrememployees VALUES (107, 'Diana', 'Lorentz', 'DLORENTZ', '590.423.5567', TO_DATE('07-FEB-2
'IT_PROG', 4200, NULL, 103, 60);
INSERT INTO hrememployees VALUES (124, 'Kevin', 'Mourgos', 'KMOURGOS', '650.123.5234', TO_DATE('16-NOV-2
'ST_MAN', 5800, NULL, 100, 50);
INSERT INTO hrememployees VALUES (141, 'Trenna', 'Rajs', 'TRAJS', '650.121.8009', TO_DATE('17-OCT-2
'ST_CLERK', 3500, NULL, 124, 50);
INSERT INTO hrememployees VALUES (149, 'Eleni', 'Zlotkey', 'EZLOTKEY', '011.44.1344.429018', TO_DATE('29
'SA_MAN', 10500, .2, 100, 80);
INSERT INTO hrememployees VALUES (174, 'Ellen', 'Abel', 'EABEL', '011.44.1644.429267', TO_DATE('11
'SA_REP', 11000, .3, 149, 80);
```

Output :

	EMPLOYEE_ID	LAST_NAME	EMAIL	SALARY	COMMISSION_PCT	HIRE_DATE
1	1	King	king@mail.com	5000	0,1	01-01-2023
2	2	Kochhar	kochhar@mail.com	4500	(null)	15-02-2023
3	3	Abel	abel@mail.com	3800	0,2	10-03-2023
4	4	Taylor	taylor@mail.com	3200	0,15	05-04-2023
5	5	Grant	grant@mail.com	2800	0,1	20-05-2023
6	6	Hunold	hunold@mail.com	4000	(null)	20-01-2023
7	7	Ernst	ernst@mail.com	3500	(null)	10-02-2023
8	8	Lorentz	lorentz@mail.com	3000	(null)	01-03-2023
9	9	Whalen	whalen@mail.com	2500	(null)	12-04-2023
10	10	Zlotkey	zlotkey@mail.com	3100	0,1	01-05-2023

Code 2

Query :

```
CREATE OR REPLACE PROCEDURE increase_salary (  
    department_id_in IN employees.department_id%TYPE,  
    increase_pct_in IN NUMBER  
) IS  
BEGIN  
    FOR employee_rec IN (  
        SELECT employee_id  
        FROM employees  
        WHERE department_id = increase_salary.department_id_in  
    ) LOOP  
        UPDATE employees emp  
        SET emp.salary = emp.salary + (emp.salary * increase_salary.increase_pct  
        WHERE emp.employee_id = employee_rec.employee_id;  
  
    END LOOP;  
END increase_salary;  
/  
  
BEGIN  
    increase_salary (90, .50);  
    ROLLBACK;  
END;  
/  
  
select * from employees where department_id = 90
```

Output :

	EMPLOYEE_ID	LAST_NAME	EMAIL		SALARY	COMMISSION_PCT	HIRE_DATE	DEPART
1	1	King	king@mail.com		5000	0,1	01-01-2023	
2	2	Kochhar	kochhar@mail.com		4500	(null)	15-02-2023	

Query Rollback dihapus

Output dibagian salary akan bertambah karena mengincrease gaji/salary
increase_salary

Query :

```
BEGIN
  increase_salary (90, .50);
END;
/
```

Output :

	EMPLOYEE_ID	LAST_NAME	EMAIL	SALARY	COMMISSION_PCT	HIRE_DATE	DEPART
1	1	King	king@mail.com	7500	0,1	01-01-2023	
2	2	Kochhar	kochhar@mail.com	6750	(null)	15-02-2023	